

Practical Questions

19 May 2026 07:47

ST. XAVIER'S COLLEGIATE SCHOOL, 30 MOTHER TERESA SARANI, KOLKATA-16
Class: XI **FIRST SEMESTER BLOCK TEST** **Date: 02/09/2024**
Duration: Three hours **COMPUTER SCIENCE PRACTICALS** **Full Marks: 30**

The total time to be spent on the Planning Session and the Examination Session is **Three hours**. After completing the Planning Session, the candidate may begin with the Examination Session. A maximum of 90 minutes is permitted for the Planning Session. However, if candidates finish earlier, they are to be permitted to begin the Examination Session.

This paper consists of three problems from which candidates are required to attempt any **one** problem. Candidates are expected to do the following:

A. Planning Session:

1. Write an algorithm for the selected problem. [3 marks]
(Algorithm should be expressed clearly using any standard scheme such as pseudo code or in steps which are simple enough to be obviously computable.)
2. Write a program in **JAVA** language. The program should follow the algorithm and should be logically and syntactically correct. Document the program using mnemonic names / comments, identifying and clearly describing the choice of data types and meaning of variables. [7 marks]

B. Examination Session:

1. Code / Type the program on the computer. Typically, this should be a program that compiles and runs correctly. [2 marks]
2. Test run the program on the computer using the given sample data and get the output in the format specified in the problem. [3 marks]

Note: The candidates must not carry any stationery, items such as pen / pencil / eraser to the Computer Laboratory for the Examination Session.

Solve any **one** of the following Problems.

Question 1:

Given two positive numbers M and N , such that M is between 100 and 10000 and N is less than 100. Find the smallest integer that is greater than M and whose digits add up to N . For example, if $M = 100$ and $N = 11$, then the smallest integer greater than 100 whose digits add up to 11 is 119.

Write a program to accept the numbers M and N from the user and print the smallest required number whose sum of all its digits is equal to N . Also, print the total number of digits present in the required number. The program should check for the validity of the inputs and display an appropriate message for an invalid input.

Test your program with the sample data and some random data:

Example 1

INPUT :

$M = 100$

$N = 11$

OUTPUT :

The required number = 119

Total number of digits = 3

Example 2

INPUT :

$M = 1500$

$N = 25$

OUTPUT :

The required number = 1699

Total number of digits = 4

Example 3

INPUT :

$M = 99$

Example 3

INPUT :

M = 99

N = 11

OUTPUT :

INVALID INPUT

Example 4

INPUT :

M = 112

N = 130

OUTPUT :

INVALID INPUT

Question 2

Write a program to declare a matrix $A[][]$ of order $(M \times N)$ where 'M' is the number of rows and 'N' is the number of columns such that the value of 'M' must be greater than 0 and less than 10 and the value of 'N' must be greater than 2 and less than 6. Allow the user to input digits (0 - 7) only at each location, such that each row represents an octal number.

Example: 2 3 1 (decimal equivalent of 1st row = 153 i.e. $2 \times 8^2 + 3 \times 8^1 + 1 \times 8^0$)

4 0 5 (decimal equivalent of 2nd row = 261 i.e. $4 \times 8^2 + 0 \times 8^1 + 5 \times 8^0$)

1 5 6 (decimal equivalent of 3rd row = 110 i.e. $1 \times 8^2 + 5 \times 8^1 + 6 \times 8^0$)

Perform the following tasks on the matrix:

(a) Display the original matrix.

(b) Calculate the decimal equivalent for each row and display as per the format given below.

Test your program for the following data and some random data:

Example 1

INPUT: M = 1

N = 3

ENTER ELEMENTS FOR ROW 1: 1 4 4

OUTPUT:	FILLED MATRIX	DECIMAL EQUIVALENT
	1 4 4	100

Example 2

INPUT: M = 3

N = 4

ENTER ELEMENTS FOR ROW 1: 1 1 3 7

ENTER ELEMENTS FOR ROW 2: 2 1 0 6

ENTER ELEMENTS FOR ROW 3: 0 2 4 5

OUTPUT:	FILLED MATRIX	DECIMAL EQUIVALENT
	1 1 3 7	607
	2 1 0 6	1094
	0 2 4 5	165

Question 3

Question 3

Write a program to accept a date in the string format dd/mm/yyyy and accept the name of the day on 1st of January of the corresponding year. Find the day for the given date.

Example:

INPUT

Date : 5/7/2024 Day on 1st January : MONDAY

OUTPUT

Day on 5/7/2024 : FRIDAY

Test run the program on the following inputs.

<u>INPUT DATE</u>	<u>DAY ON 1ST JANUARY</u>	<u>OUTPUT DAY FOR INPUT DATE</u>
4/9/1998	THURSDAY	FRIDAY
31/8/1999	FRIDAY	TUESDAY
6/12/2000	SATURDAY	WEDNESDAY

The program should include the part for validating the inputs namely the date.